Lesson 3.2: Introduction to Mixed Actions

Mixed actions

Victor Jauregui

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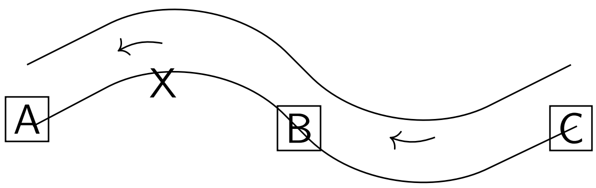
River delivery



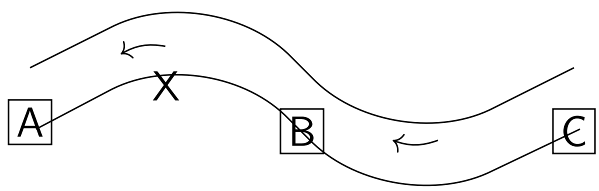
Example (River delivery)

Alice delivers goods to town C from her warehouse at X on a river that flows from C to A. Deliveries are made using motor boats which consume fuel only when travelling against the current. On some days, a goods ferry (f) travels up the river, stopping at A, then B, and finally C, but not at X. The ferry will tow boats up the river for free on the days it operates.

Fuel consumption (litres):

	A	X	B	C
To C from	4	3	2	0

River delivery (continued)



Delivery alternatives to C (from X):

- A via A (floating down river)
- B via B (travelling first to B)
- C travelling directly to C

Fuel consumed:

	$\overline{f}$	f
A	4	0
B	3	1
C	3	3

Multiple deliveries

Regret: extra fuel used

	$\overline{f}$	f
A	1	0
B	0	1

Four deliveries, same day:

	$\overline{f}$	f
A	1	0
$\frac{3}{4}$A$\frac{1}{4}$B	$\frac{3}{4}$	$\frac{1}{4}$
$\frac{1}{2}$A$\frac{1}{2}$B	$\frac{1}{2}$	$\frac{1}{2}$
$\frac{1}{4}$A$\frac{3}{4}$B	$\frac{1}{4}$	$\frac{3}{4}$
B	0	1